

New Asymptotic Results on Predicting Non-linear Polynomial Congruential Generators

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Abstract. We investigate cryptanalytic attacks on non-linear polynomial congruential generators (PCGs), a class of number-theoretic pseudorandom number generators. A PCG operates by iterating an algebraic map $F(x) \bmod p$ on a secret initial seed v_0 using fixed parameters, and outputs a truncated portion of each state (for example, the most significant bits). We propose new and improved lattice-based attacks that exploit systems of modular polynomial equations derived from PCGs.

Specifically, we analyze three common non-linear PCGs: the Quadratic Congruential Generator (QCG), the Power Generator, and the Pollard Generator. We establish asymptotic bounds for predicting these PCGs, assuming the adversary has access to an infinitely long output sequence. To derive these bounds, we develop new symbolic techniques that build on the automated Coppersmith’s method framework recently developed by Feng et al. (Crypto ’25). Our approach is more flexible than previous methods and is particularly well-suited for deriving symbolic bounds. Applying our techniques, we obtain the best-known analytical results for asymptotic attacks on these PCGs:

- We present, for the first time, asymptotic attack bounds on QCGs with partially known coefficients.
- We extend and improve the asymptotic attack of Herrmann and May (Asiacrypt ’09) on Power Generators.
- We improve the asymptotic attack of Bauer et al. (PKC ’12) on Pollard Generators and confirm their conjecture.

We validate our theoretical findings with numerical experiments that demonstrate the practicality and efficacy of our attacks.

Keywords: Asymptotic bound · Attack · Automated Coppersmith’s method · Lattice reduction · Polynomial congruential generator

1 Introduction

Pseudorandom number generators (PRNGs) are a fundamental primitive in modern cryptography, essential for creating secret keys, initialization vectors, and nonces. A PRNG is a deterministic algorithm that, given a secret initial value known as a seed, produces a sequence of numbers that is computationally indistinguishable from a truly random sequence. The core of a PRNG is an iterative function F applied over a finite field or ring. Starting with a seed v_0 , the sequence is generated by the recurrence $v_{i+1} = F(v_i) \bmod p$, where a portion of each state v_i is output at each step. The security of such a generator, therefore, hinges on the resilience of the function F against attacks that attempt to predict its output based on partial information.

The simplest and most well-known class of PRNGs is the Linear Congruential Generator (LCG) [12], defined by the affine function $F(x) = ax + b$. Although LCGs are efficient and exhibit acceptable statistical properties for some applications, they are cryptographically insecure. As Boyar [7] demonstrated, even when only a fraction of the output is revealed, such as a few of the most significant bits (MSBs), it is possible to recover the generator's parameters and seed in polynomial time.

This inherent weakness [8,19,25,28] in linear constructions led researchers to explore non-linear functions to enhance security. However, the introduction of non-linearity is not a panacea. A significant body of research [1,3,5,13,14,15,16] has shown that many non-linear PRNGs remain vulnerable, especially when a sufficient number of output bits are known. These attacks often employ sophisticated algebraic and lattice-based techniques to reconstruct the internal state of the generator and predict its future outputs. A classic example is the analysis of the Blum-Blum-Shub (BBS) generator [6], which is based on the quadratic residue recurrence $v_{i+1} = v_i^e \bmod N$ for a positive integer e and a random seed v_0 , where $N = pq$ is the product of two large secret prime numbers. Herrmann and May [16] showed that if a significant portion, i.e., a $(1 - 1/e)$ fraction of the most significant bits of each output is known, the entire sequence can be recovered.

This work focuses on Polynomial Congruential Generators (PCGs), which are defined by a polynomial recurrence of degree $d \geq 2$:

$$v_{i+1} = F(v_i) \bmod p, \text{ where } F(x) = a_d x^d + \cdots + a_1 x + a_0.$$

This family includes the Quadratic Congruential Generator (QCG), where $F(x) = ax^2 + b$, as a special case for $d = 2$. Another notable example is the Power Generator, defined by $v_{i+1} = v_i^e \bmod p$ for some integer exponent $e \geq 2$. The BBS generator is a specific instance of this family where the modulus is a composite number N . Another important PCG is the Pollard Generator, defined by the recurrence $F(x) = x^2 + c$, which is closely related to the QCG.

Our objective is to analyze the predictability of these generators, particularly when an attacker has access to truncated outputs. The central question is to determine the threshold of information output: how many bits of each output

must be revealed for the generator's secret state and parameters to become recoverable.

In a cryptographic setting, the initial value v_0 (and sometimes the polynomial F) is assumed to be secret, and the outputs of the generator are used as a keystream. In most settings, only the most significant bits of each v_j are used, with the hope that this makes the resulting output sequence more difficult to predict. A central research question is how many bits of each state v_j can be safely output at each iteration. If too many bits are revealed, the generator becomes insecure, as an attacker can efficiently recover the sequence of states v_j . More precisely, our goal is to determine the minimum number of bits from consecutive outputs that must be revealed for the generator to be predictable in polynomial time in $\log p$.

The core observation is that there is an underlying algebraic dependency between consecutive states, a weakness that lattice-based attacks exploit. To illustrate, consider a general PCG. We have the following equations:

$$\begin{aligned} v_{i+1} &\equiv a_d v_i^d + \cdots + a_1 v_i + a_0 \pmod{p}, \\ v_{i+2} &\equiv a_d v_{i+1}^d + \cdots + a_1 v_{i+1} + a_0 \pmod{p}, \\ &\vdots \\ v_{i+d+1} &\equiv a_d v_{i+d}^d + \cdots + a_1 v_{i+d} + a_0 \pmod{p}. \end{aligned}$$

The simplest case is when F is known and approximations of two consecutive states v_i and v_{i+1} are given. The goal is to recover the unknown parts u_i, u_{i+1} from the relation

$$v_{i+1} \equiv a_d v_i^d + \cdots + a_1 v_i + a_0 \pmod{p},$$

where $v_j = w_j + u_j$ with known MSBs w_j and unknown LSBs u_j . Substitution and rearrangement yield the following relation:

$$a_d(x_1 + w_i)^d + \cdots + a_1(x_1 + w_i) + a_0 - (x_2 + w_{i+1}) \equiv 0 \pmod{p}. \quad (1)$$

We can make this polynomial monic in x_1 by multiplying by $a_d^{-1} \pmod{p}$. This leads to a new modular polynomial:

$$f(x_1, x_2) = x_1^d + c_1 x_1^{d-1} + \cdots + c_{d-1} x_1 + c_d x_2 + c_0 \pmod{p},$$

where the coefficients c_i are all known. Solving such equations to predict QCGs, Power Generators, and Pollard Generators is a well-studied problem, and several strong results have been achieved.

However, several open problems remain in the cryptanalysis of PCGs. The first is *flexibility*, which aims to relax the restriction on the truncated position of PCG outputs. In most previous attacks, the PCG outputs are assumed to be the MSBs of the map function F at each iteration. To remove this restriction, one shall apply a transformation to the derived equations before using them in a lattice-based attack. This allows each state to be expressed with a single

unknown part, regardless of whether the known bits are MSBs, LSBs, or a mix of both.

The second is *extensibility*, which concerns predicting PCGs or recovering the map function F when its parameters are not fully known. As the number of unknowns increases, one needs more consecutive outputs to mount a successful attack. Specifically, recovering t unknown coefficients in F requires $t+2$ consecutive outputs. One must first apply resultants or a similar technique to eliminate the unknown coefficients from the system of $t+1$ state-transition equations. A new modular polynomial with known coefficients is then derived using the transformation described above.

The third is *asymptoticity*, which aims to obtain asymptotic attack bounds for predicting PCGs, assuming an attacker has access to an infinite number of outputs. The goal is a complete theoretical understanding that determines the ultimate security limits of the target PCGs. In this case, one must derive a system of modular polynomials, not just a single one. Thus, one proceeds to solve a system of multivariate modular equations to extract the hidden coefficients of F and fully recover the structure of the PCG.

1.1 Our Contribution

In this paper, we present new and improved asymptotic attacks against PCGs, including QCGs, Power Generators, and Pollard Generators, for scenarios where the parameters of their map function are not fully known. We demonstrate that, given a sufficient number of output bits from several consecutive values v_i , one can recover the initial seed v_0 , the map F , and thus the entire sequence.

Our attacks rely on Coppersmith’s lattice-based techniques for solving multivariate modular polynomial equations [9]. Specifically, we leverage the automated Coppersmith’s method developed by Meers and Nowakowski [24], Ryan [26], and Feng et al. [11]. Leveraging this recent tool, we can provide asymptotic analysis bounds. While previous tools could compute numerical bounds for finding small roots, they could not be directly applied to obtain *symbolic* bounds for a system of n modular polynomial equations. To overcome this obstacle, we present innovative techniques for computing the associated Newton polytopes for a symbolic system of modular equations.

Table 1 summarizes the best-known asymptotic bounds on common PCGs, including existing ones and our new results. For each PCG and parameter setting, it shows the proportion of unknown bits that may remain hidden while still enabling a (heuristic) polynomial-time attack. The “asymptotic” proportion assumes an infinite number of outputs.

Our asymptotic bounds, obtained even with partial or no knowledge of the map function, offer improved insight into the security of several common PCGs. The theoretical implications suggest that securely deploying PCGs would require outputting fewer bits per iteration, thus degrading their efficiency. The main contributions of this paper are as follows:

- Our method improves the tool developed by Feng et al. [11] to express the exponents arising in automated Coppersmith’s method in symbolic form, as

Table 1. State-of-the-art asymptotic bounds for attacks on common PCGs

PCG Type	Map $F(x)$	Attack Scenario	Previous Bound	Our Bound
QCG	$ax^2 + b$	a, b known	1/2 ([1])	1/2 (Appendix A.1)
	$ax^2 + b$	a known, b unknown	–	1/2 (Section 3.1)
	$ax^2 + b$	a unknown, b known	–	1/3 * (Section 3.2)
	$ax^2 + b$	a, b unknown	1/3 ([1])	1/3 (Appendix A.2)
Power Generator	x^e	–	1/e ([16])	1/e (Section 4)
Perturbed Power Generator	$x^e + c$	c unknown	–	1/e (Section 4)
Pollard Generator	$x^2 + c$	c known	1/2 ([1])	1/2 (Appendix A.3)
	$x^2 + c$	c unknown	2/5 ([1])	1/2 (Section 5)

* A knowledgeable reader may notice that the 1/3 bound for the case with unknown a and known b can be directly inferred from the case where both a and b are unknown. However, our point here is that, within the framework of Coppersmith’s method, the bound remains 1/3 even when b is known.

functions of a parameter n related to the number of equations. This contrasts with the numerical interpolation approaches of Meers and Nowakowski [24] and Ryan [26], and broadens the applicability of the underlying automated framework.

- Our asymptotic bound of 1/2 for QCGs with known a and unknown b is a new result. Similarly, our bound of 1/3 for QCGs with unknown a and known b is new. Comparing the four attack scenarios on QCGs, we observe that knowledge of b provides no asymptotic advantage: we achieve a bound of 1/2 for known a regardless of whether b is known, and a bound of 1/3 for unknown a regardless of whether b is known.
- We achieve an asymptotic bound of 1/e for Power Generators, matching the result of Herrmann and May [16] for the specific case of truncated MSB outputs. They used a technique known as *Unravelled Linearization*, which provides a tighter bound once a specific value of n is substituted, but at the cost of a cumbersome n -fold summation. In contrast, we show that the bound 1/e can be achieved without relying on Unravelled Linearization. By establishing the relation between the bound and its corresponding polytope, we avoid this n -fold summation through a geometric formulation within a unified framework consistent with that used for QCGs. Moreover, our analysis is more general, covering flexible attack scenarios where the truncated output may appear in arbitrary positions, leaving only a single continuous block of bits unknown. We further show that the same bound of 1/e can be achieved even when the power generator polynomial $F(x) = x^e$ is perturbed by an unknown masking constant c .
- Our asymptotic bound of 1/2 for Pollard Generators with unknown c is a new result. It improves upon the previous bound 2/5 given by Bauer et al. [1]. In their work, the authors conjectured that the bound could be improved to 1/2, our result can thus be viewed as an analogous proof of this conjecture. Since the asymptotic bound for Pollard Generators with known c is also 1/2,

this implies that knowing c brings no asymptotic advantage, an observation similar to our findings for QCGs.

- We complement our asymptotic results with experiments using practical moderate parameters to verify the correctness and effectiveness of our attacks.

In addition, Bauer et al. [1, Remark 1] conjectured that there exists a general asymptotic bound of $1/D$ for solving a system of polynomial equations of degree D derived from a map function F with truncated outputs. Our proposed attacks provide strong evidence for this conjecture.

1.2 Technical Overview

We now give a detailed overview of the main techniques used to obtain our results. Our approach utilizes preprocessing of derived modular polynomials, the lattice-based solving strategy, and innovative symbolic computation techniques to build a framework for deriving asymptotic bounds on common PCGs under various attack scenarios.

Symbolic Analysis. Prior works, such as Meers and Nowakowski [24] and Ryan [26], provide tools to automate the computation of Coppersmith’s method bounds. However, these tools are primarily numerical: given a concrete polynomial system (e.g., for $n = 2$ or $n = 3$), they can compute the specific attack bound (e.g., $\delta_{(2)}$ or $\delta_{(3)}$), but they cannot derive a general, symbolic expression $\delta_{(n)}$ in terms of n . The work of Feng et al. [11] connects this bound to the volumes of associated Newton polytopes and provides theoretical formulas for its computation. But the tool in [11] is still oriented towards concrete polynomial systems rather than symbolic systems parameterized by n .

A key technical contribution of our work is solving this symbolic computation challenge. We observe that the systems of equations derived from several PCGs have a special structure. For instance, given $n + 1$ consecutive outputs, we may obtain a system of $n - 1$ equations in $n + 1$ variables. We show that this system corresponds to a class of special polytopes whose dimension and structure depend on the symbolic parameter n . However, computing the volume of these parameterized polytopes remains nontrivial, as they have dimension $n + 2$ for $n \geq 2$, and even a four-dimensional polytope cannot be easily visualized. To address this challenge, we develop several techniques that enable symbolic evaluation of their volumes. Our approach begins by projecting the irregular $(n + 2)$ -dimensional polytope onto an n -dimensional subspace, resulting in a more regular polytope with several missing corners. We then complete these missing corners to facilitate computation and subsequently subtract their contribution. Ultimately, we obtain a closed-form expression for the general attack bound $\delta_{(n)}$.

Flexibility. We express each state v_j at the j -th iteration as $v_j = w_j + 2^{l_j}u_j$, where u_j is the single unknown part and w_j is the known part of the output.

Additionally, l_j indicates the number of known LSBs, and it can differ for each iteration, which implies that we do not strictly require the PRNG to have a fixed output pattern. Taking $v_{i+1} = F(v_i)$ with known F as an example, we obtain a polynomial that differs from the standard form in (1):

$$a_d(2^{l_i}x_1 + w_i)^d + \cdots + a_1(2^{l_i}x_1 + w_i) + a_0 - (2^{l_{i+1}}x_2 + w_{i+1}) \equiv 0 \pmod{p}.$$

Multiplying by the inverse $a_d^{-1}2^{-l_id} \pmod{p}$ results in a modular polynomial $f(x_1, x_2)$ that is monic in x_1 :

$$f(x_1, x_2) = x_1^d + c_1x_1^{d-1} + c_2x_1^{d-2} + \cdots + c_{d-1}x_1 + c_dx_2 + c_{d+1}.$$

This preprocessing simplifies subsequent calculations in Coppersmith's method and the symbolic derivation of asymptotic bounds. For Power Generators, this preprocessing always yields a modular polynomial of the form:

$$f_{\text{PowerG}}(x_1, x_2) = x_1^e + c_1x_1^{e-1} + \cdots + c_{e-1}x_1 + c_ex_2 + c_{e+1}. \quad (2)$$

Extensibility. When the coefficients of F are not fully known, we use techniques such as resultants to eliminate them from a system of equations derived from consecutive outputs v_j . For a general PCG of a given degree d with some known coefficients a_j , this elimination can be performed algorithmically. For the QCGs, Power Generators, and Pollard Generators studied in this paper, we can manually derive the respective modular polynomials.

We require three consecutive outputs for a QCG where one coefficient (a or b) is unknown, yielding equations $v_{i+1} = av_i^2 + b$ and $v_{i+2} = av_{i+1}^2 + b$. From these, we obtain $f_{\text{QCG1}}(x_1, x_2, x_3)$ for the case of known a and unknown b :

$$f_{\text{QCG1}}(x_1, x_2, x_3) = x_1^2 + c_1x_2^2 + c_2x_1 + c_3x_2 + c_4x_3 + c_0. \quad (3)$$

Similarly, we obtain $f_{\text{QCG2}}(x_1, x_2, x_3)$ for the case of unknown a and known b :

$$f_{\text{QCG2}}(x_1, x_2, x_3) = x_2^3 + c_1x_1^2x_3 + c_2x_1^2 + c_3x_2^2 + c_4x_1x_3 + c_5x_1 + c_6x_2 + c_7x_3 + c_0. \quad (4)$$

For a Pollard Generator using $F(x) = x^2 + c$ with an unknown parameter c , we deduce the modular polynomial $f_{\text{PollardG}}(x_1, x_2, x_3)$ below:

$$f_{\text{PollardG}}(x_1, x_2, x_3) = x_1^2 + c_1x_2^2 + c_2x_1 + c_3x_2 + c_4x_3 + c_0. \quad (5)$$

Asymptoticity. To obtain asymptotic attack bounds, we assume an attacker has access to an infinite output sequence and study the limit behavior by analyzing an attack using $n+1$ consecutive outputs, where n is a parameter. These $n+1$ outputs yield n algebraic relations. After eliminating unknown parameters, this typically results in a system of $n-1$ modular polynomial equations. For example, in the QCG case (with known a , unknown b), this requires solving

a system of the form:

$$\begin{cases} f_1(x_1, x_2, x_3) \equiv 0 \pmod{p}, \\ f_2(x_2, x_3, x_4) \equiv 0 \pmod{p}, \\ \vdots \\ f_{n-1}(x_{n-1}, x_n, x_{n+1}) \equiv 0 \pmod{p}, \end{cases}$$

where each polynomial f_j for $j = 1, \dots, n-1$ is an analogue of (3).

How to solve such a symbolic system of equations, parameterized by n , is the most central technical challenge of this work. We apply our symbolic analysis framework, combined with innovative polytope volume computation techniques, to handle this system. This allows us to derive a general attack bound in terms of n and, by taking the limit as $n \rightarrow \infty$, to determine the final asymptotic security threshold.

1.3 Related Work

The predictability of non-linear PRNGs has been extensively studied, particularly under the assumption that an adversary has access to truncated outputs. The core technique behind most attacks is the application of lattice-based methods, particularly Coppersmith's method for finding small roots of modular polynomial equations [9]. This approach reduces the problem of recovering secret states or parameters to solving a system of congruences, where the unknown bits of the output are treated as small variables.

QCGs. The predictability of QCGs, defined by the recurrence $v_{i+1} = av_i^2 + b \pmod{p}$, has been analyzed under various assumptions regarding which parameters (a or b) are known. For instance, if a is known, such attacks proceed by observing three consecutive outputs v_i, v_{i+1}, v_{i+2} and eliminating b to obtain an equation $v_{i+1} - v_{i+2} = a(v_i^2 - v_{i+1}^2) \pmod{p}$. By splitting each value into a known part (from the output) and an unknown part, this equation becomes a multivariate polynomial in the unknowns, which can then be solved.

Blackburn et al. [4] provided foundational results for the QCG, considering cases where both a and b are known and where both are unknown. In the known-parameter scenario, they showed that recovering the state is feasible if at most $1/4$ of the bits per output are unknown. When both parameters are unknown, this threshold reduces to $1/19$. Bauer et al. [1] improved these bounds: if both a and b are known, up to $1/3$ of the output may remain hidden; if both are unknown, the limit becomes $1/12$. They also provided asymptotic results, assuming an infinite number of outputs: $1/2$ for known a, b and $1/3$ for unknown a, b .

Gómez et al. [14] examined the case where a is known and b is unknown, showing predictability when less than $1/5$ of each output is hidden. More recently, Gao et al. [13] improved this bound to $1/4$ by restructuring the polynomial equations using variable separation. However, open questions remained, such as whether these attacks could be enhanced using more than three outputs.

Power Generators. The Power Generator, defined by $v_{i+1} = v_i^e \bmod N$, represents a different class of non-linear generators. Attacks against this generator [5,16,27] are similar to those against PCGs and typically focus on output from the most significant bits. The public exponent e does not need to be eliminated.

The work by Herrmann and May [16] provides a powerful attack based on Coppersmith’s method combined with a technique called unraveled linearization. The attack starts with the relation $v_{i+1} = v_i^e \bmod N$. The attacker is assumed to know an approximation w_i of a state v_i , such that $v_i = w_i + u_i$, where u_i is the small unknown part. For $e = 2$, observing two iterations v_1, v_2 gives the congruence $w_2 + u_2 = (w_1 + u_1)^2 \bmod N$. This can be rewritten as a bivariate polynomial $f(x_1, x_2) = x_1^2 - x_2 + 2w_1x_1 + w_1^2 - w_2 = 0 \bmod N$ in two small variables $(x_1, x_2) = (u_1, u_2)$. The upper bound for x_1, x_2 is $N^{2/5}$. However, their results were restricted to MSB outputs, leaving the case of LSB outputs as an open problem.

Herrmann and May generalized their approach to an arbitrary number of iterations, allowing them to investigate the asymptotic behavior. For a polynomial of degree e , their method finds roots up to size $N^{1/e}$. The attack succeeds if the unknown part is smaller than this bound, which corresponds to revealing a $(1 - 1/e)$ fraction of the MSBs.

Pollard Generators. The Pollard Generator, defined by $v_{i+1} = v_i^2 + c \bmod p$, is a simplified variant of the QCG. Blackburn et al. [4] established that the generator is predictable if an attacker has access to good approximations of two consecutive values, specifically if at least $2/3$ of the MSBs of each output are revealed. They also showed that with an unknown shift c , the generator is insecure if at most $1/4$ of each output is unknown.

An improvement by Gómez et al. [15] showed the generator is vulnerable even when at most $5/14$ of the LSBs are unknown. Their method uses an iterative application of the Closest Vector Problem (CVP). They noted their approach could be extended to an unknown shift c and that the asymptotic bound on unknown LSBs could reach $2/5$.

Later, Bauer et al. [1] improved these bounds further: if c is known, up to $2/5$ of the output may remain hidden; if c is unknown, the limit is $2/7$. They also provided asymptotic results, assuming an infinite number of outputs: $1/2$ for known c and $2/5$ for unknown c .

1.4 Organization

The remainder of this paper is structured as follows. Section 2 reviews the necessary background on lattice-based techniques. Section 3 details our asymptotic attacks on QCGs with partially known coefficients. The asymptotic attacks on Power Generators and Pollard Generators are presented in Sections 4 and 5, respectively. Finally, Section 6 concludes the paper with summary remarks.

2 Preliminaries

This section recalls the key concepts and techniques underpinning our lattice-based attacks. We review the LLL lattice reduction algorithm [21], Copper-smith’s method [9] including the refinement by Howgrave-Graham [17], and recent advancements by Feng et al. [11]. For more comprehensive treatments, we refer readers to [22,23,30].

2.1 Polynomials

Let $f \in \mathbb{Z}[x_1, \dots, x_k]$ be a polynomial in k variables, written as

$$f(x_1, \dots, x_k) = \sum_{i_1, \dots, i_k \in \mathbb{N}} \alpha_{i_1, \dots, i_k} x_1^{i_1} \cdots x_k^{i_k},$$

where $\alpha_{i_1, \dots, i_k} \in \mathbb{Z}$. A term $x_1^{i_1} \cdots x_k^{i_k}$ with $\alpha_{i_1, \dots, i_k} \neq 0$ is called a *monomial* of f .

A *monomial order* $\prec$ on the set of monomials $\mathcal{M}$ is a total order satisfying that $1 \prec \lambda$ for all $\lambda \in \mathcal{M}$, and if $\lambda_1 \prec \lambda_2$, then $\lambda\lambda_1 \prec \lambda\lambda_2$ for all $\lambda \in \mathcal{M}$. Given $\prec$, the *leading monomial* $\text{LM}(f)$ is the maximal monomial of f under $\prec$, and the *leading coefficient* $\text{LC}(f)$ is its coefficient.

We assume $x_1 \succ \cdots \succ x_k$ and use two standard monomial orders in our paper. The lexicographic order ($\prec_{\text{lex}}$) compares exponents of x_1 , then x_2 , and so on. For example, we have $1 \prec x_2 \prec x_2^2 \prec x_1 \prec x_1x_2 \prec x_1^2$. And the graded reverse lexicographic order compares total degrees first, and then reverses the lexicographic comparison. For example, we have $x_3^2 \prec x_2x_3 \prec x_1x_3 \prec x_2^2 \prec x_1x_2 \prec x_1^2$.

2.2 Lattices and Coppersmith’s Method

Let n and ω be positive integers with $\omega \leq n$. Let $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_\omega$ in $\mathbb{R}^n$ be ω linearly independent vectors. The *lattice* Λ spanned by $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_\omega$ is formally defined as the set of all their integer linear combinations. That is

$$\Lambda = \left\{ \sum_{i=1}^{\omega} z_i \mathbf{b}_i \mid z_i \in \mathbb{Z} \right\}.$$

The *determinant* of the lattice, denoted $\det(\Lambda)$, is given by $\sqrt{\det(BB^\top)}$, where B is the matrix with rows $\mathbf{b}_i$. When the lattice has full rank ($\omega = n$), the determinant simplifies to $\det(\Lambda) = |\det(B)|$.

The celebrated LLL algorithm [21] provides an efficient way to compute a reduced basis of a lattice. The following lemma captures the bound on the vector lengths in the reduced basis.

Lemma 1. *Let Λ be an ω -dimensional lattice generated by $\mathbf{b}_1, \dots, \mathbf{b}_\omega$. The LLL algorithm returns a reduced basis $(\mathbf{v}_1, \dots, \mathbf{v}_\omega)$ satisfying*

$$\|\mathbf{v}_i\| \leq 2^{\frac{\omega(\omega-1)}{4(\omega+1-i)}} \cdot \det(\Lambda)^{\frac{1}{\omega+1-i}}, \text{ for } i = 1, \dots, \omega.$$

The algorithm runs in time polynomial in ω and in the logarithm of the maximum absolute entry in the original basis.

Howgrave-Graham's criterion [17] makes it possible to check whether a root of a polynomial modulo R is also a root over the integers. Consider an integer polynomial

$$h(x_1, \dots, x_n) = \sum c_{i_1, \dots, i_n} x_1^{i_1} \cdots x_n^{i_n},$$

with its Euclidean norm defined by

$$\|h\| = \sqrt{\sum |c_{i_1, \dots, i_n}|^2}.$$

Lemma 2. *Let $h(x_1, \dots, x_n) \in \mathbb{Z}[x_1, \dots, x_n]$ be a polynomial with at most ω monomials. Let R be a modulus, and let $X_1, \dots, X_n$ be bounds on the absolute values of the variables. If there exists a solution $(x_1^*, \dots, x_n^*)$ with $|x_i^*| \leq X_i$ such that $h(x_1^*, \dots, x_n^*) \equiv 0 \pmod{R}$ and $\|h(X_1 x_1, \dots, X_n x_n)\| < R/\sqrt{\omega}$, then $h(x_1^*, \dots, x_n^*) = 0$ over the integers.*

Combining Lemma 1 and Lemma 2 yields a powerful approach. Suppose we compute the first ℓ reduced vectors $(\mathbf{v}_1, \dots, \mathbf{v}_\ell)$ of a lattice Λ of dimension ω . To ensure that one of the associated polynomials vanishes at the solution over $\mathbb{Z}$, it suffices to satisfy

$$2^{\frac{\omega(\omega-1)}{4(\omega+1-\ell)}} \cdot \det(\Lambda)^{\frac{1}{\omega+1-\ell}} < \frac{R}{\sqrt{\omega}}.$$

For asymptotic analysis using lattices of large dimension, this is often simplified to

$$\det(\Lambda) < R^{\omega-\varepsilon}, \tag{6}$$

where ε denotes a term that is negligible compared to ω .

The workflow of a lattice-based attack is as follows: 1) Let $f(x_1, \dots, x_n)$ be a polynomial such that $f(x_1^*, \dots, x_n^*) \equiv 0 \pmod{M}$ with bounds $|x_i^*| \leq X_i$. 2) Choose an integer m and construct shift polynomials satisfying $\mathfrak{s}_k(x_1^*, \dots, x_n^*) \equiv 0 \pmod{M^m}$. A typical form is $x_1^{i_{k1}} \cdots x_n^{i_{kn}} f^{j_k} M^{m-j_k}$. 3) Scale each polynomial $\mathfrak{s}_k$ by the bounds X_i and form a matrix B from their coefficient vectors. This matrix generates a lattice Λ . 4) Apply the LLL algorithm to reduce B , producing short vectors $\mathbf{v}_1, \dots, \mathbf{v}_\ell$, which correspond to new polynomials that vanish at $(x_1^*, \dots, x_n^*)$ under certain conditions. 5) Assuming the new polynomials are algebraically independent, compute the solution using resultants or a Gröbner basis computation [2]. The time complexity of Gröbner basis computation is, in general, doubly exponential in the degree of the polynomials.

Note that Coppersmith's method for multivariate polynomials is heuristic, as there is no formal guarantee that the constructed polynomials are algebraically independent. Nevertheless, this assumption holds in most practical scenarios and is widely accepted in cryptanalytic literature. We adopt this assumption throughout our paper.

Assumption 1 *The lattice-based Coppersmith's method yields algebraically independent polynomials. The common roots of these polynomials can be computed efficiently using Gröbner basis computation.*

2.3 Automated Computing Tool

An effective lattice basis construction is essential for success. For solving a single polynomial equation, Jochemsz and May [18] proposed a systematic strategy for selecting shift polynomials. Their approach yields a triangular basis matrix under a suitable monomial ordering, allowing for a simple computation of $\det(A)$ as the product of the diagonal entries.

More recently, the automation of Coppersmith's method was investigated by Meers and Nowakowski [24] and Ryan [26], who also considered systems of polynomial equations. A more recent development by Feng et al. [10] leverages additive combinatorics to derive asymptotic bounds. Their tool avoids tedious polynomial expansions and ties the success condition to the geometry of the Newton polytope of the polynomial system. The Newton polytope $A(f)$ associated with $f(x_1, \dots, x_n)$ is defined by

$$A(f) = \{(i_1, i_2, \dots, i_n) \mid x_1^{i_1} x_2^{i_2} \cdots x_n^{i_n} \text{ is a monomial of } f(x_1, \dots, x_n)\}.$$

The leading monomial of f plays an important role in the analysis. The set $A(f)$ has the structure as a sublattice of $\mathbb{Z}^n$, and its determinant is denoted by $\text{ind } A(f)$.

The work of Feng et al. revisits the Jochemsz-May strategies and extends them to systems of polynomial equations. For their basic strategy, the asymptotic success condition becomes

$$\prod_{i=1}^n X_i^{\int_{N(f)} x_i \, dV} \cdot M^{\frac{n}{n+1} \int_{N(f)} 1 \, dV} < M^{\int_{N(f)} 1 \, dV - \varepsilon},$$

where dV is the Lebesgue measure, and $N(f)$ is the convex hull of the points in $A(f)$. Specifically, $N(f)$ is defined as

$$N(f) = \left\{ \sum_{i=1}^n \lambda_i k_i \mid \lambda_i \geq 0, \sum_{i=1}^n \lambda_i = 1 \right\},$$

where $\{k_1, k_2, \dots, k_n\}$ is a set of points of $A(f)$. Moreover, they observe that the lattice dimension used in Coppersmith's method equals the size of $mA(f)$, i.e.,

the sum of m copies of $A(f)$. To compute the asymptotic bound, one needs to evaluate the limit

$$\lim_{m \rightarrow \infty} \frac{|mA(f)|}{m^n}.$$

Their analysis relies on results by Khovanskii [20] to relate sums over integer points in scaled polytopes to integrals over the polytopes themselves, as formalized in the following lemmas.

Lemma 3 (Corollary 2, [20]). *Suppose $\mathbf{0} \in A \subseteq \mathbb{Z}^k$ and A has finite index $\text{ind } A$ in $\mathbb{Z}^k$. Then there exists M such that for all sufficiently large m , the number of integer points in the scaled polytope mA is a polynomial in m of degree k . The leading coefficient equals the integral of 1 over $N(A)$ divided by $\text{ind } A$, given by*

$$\lim_{m \rightarrow \infty} \frac{|mA|}{m^k} = \frac{\int_{N(A)} 1 \, dV}{\text{ind } A}.$$

For the lattice determinant, it is further necessary to compute sums of homogeneous polynomials ϕ , such as the projection polynomial ϕ_{j_0} :

$$\begin{aligned} \phi_{j_0} : \mathbb{Z}^k &\longrightarrow \mathbb{Z}, \\ (i_1, \dots, i_k) &\longmapsto i_{j_0}. \end{aligned}$$

For such sums, the following lemma is applied.

Lemma 4 (Lemma 10, [11]). *Suppose $\mathbf{0} \in A \subseteq \mathbb{Z}^k$ and A has finite index $\text{ind } A$ in $\mathbb{Z}^k$. Then there exists M such that for all sufficiently large $m > M$, the sum*

$$\sum_{(i_1, \dots, i_k) \in mN(A) \cap \mathbb{Z}^k} i_{j_0}$$

is a polynomial in m of degree $k + 1$. The leading coefficient equals the integral of x_{j_0} over $N(A)$ divided by $\text{ind } A$, given by

$$\lim_{m \rightarrow \infty} \frac{\sum_{(i_1, \dots, i_k) \in mA} i_{j_0}}{m^{k+1}} = \frac{\int_{N(A)} x_{j_0} \, dV}{\text{ind } A}.$$

In [11], Feng et al. also consider the Jochemsz-May Extended Strategy and generalize it to systems of polynomial equations. Further details are provided in [10]. An open-source implementation for computing these asymptotic bounds is available at <https://github.com/ffmath/AsymptoticBounds>. We refer to the approach without additional shifts as the *Basic Strategy*. For many polynomial systems, it is advantageous to include additional shifts in one of the variables; we refer to this as the *Extended Strategy*.

Remark 1. We investigate asymptotic results on predicting typical PCGs based on the asymptotic solvability condition of Coppersmith's method. To avoid redundancy, we will use the term "asymptotic bound" to describe our main results and omit the term ε for simplicity.

2.4 Experimental Environment

To demonstrate the correctness and effectiveness of our proposed attacks on PCGs, we performed a series of numerical experiments with randomly generated parameters for small values of n . All experiments were carried out using SageMath [29] on a machine running 64-bit Windows 10 with Ubuntu 22.04 on WSL 2, equipped with a 2.80 GHz CPU and 16 GB of RAM.

3 Attacks on Quadratic Congruential Generators

Let p be a prime number of bit-length π . A QCG is defined by the recurrence $v_{i+1} = F(v_i) \bmod p$, where $F(x) = ax^2 + b$ is a quadratic polynomial over $\mathbb{Z}_p$, and v_0 is the secret seed.

We assume that the generator reveals κ truncated bits of each state v_i , with a known output pattern. This implies that the internal state can be written as

$$v_i = w_i + L_i u_i,$$

where w_i is the known partial output, L_i is a power of 2, and $u_i \leq p^\delta$ is the unknown portion. Our goal is to recover the sequence u_i , assuming access to infinitely many consecutive outputs and that either a or b is known, and to maximize the admissible value of δ .

3.1 The Case of Known a and Unknown b

We consider an attack using $n + 1$ consecutive outputs. This yields n relations $v_j = av_{j-1}^2 + b \bmod p$ for $j = 1, \dots, n$. Subtracting the j -th and $(j + 1)$ -th relations for $j = 1, \dots, n - 1$ eliminates b :

$$v_j - v_{j+1} \equiv a(v_{j-1}^2 - v_j^2) \bmod p.$$

Substituting $v_k = w_k + L_k u_k$ for $k = j - 1, j, j + 1$, multiplying by $a^{-1}L_{j-1}^{-2} \bmod p$, and rearranging yields

$$\begin{aligned} & a^{-1}L_{j-1}^{-2}(w_j - w_{j+1} + L_j u_j - L_{j+1} u_{j+1}) \\ & \equiv L_{j-1}^{-2}(w_{j-1}^2 - w_j^2) + 2L_{j-1}^{-1}w_{j-1}u_{j-1} + u_{j-1}^2 - 2L_{j-1}^{-2}L_j w_j u_j - L_{j-1}^{-2}L_j^2 u_j^2 \bmod p. \end{aligned}$$

Letting $x_k = u_{k-1}$ for $k = 1, \dots, n + 1$, we obtain for each $j = 1, \dots, n - 1$ a modular polynomial equation

$$f_j(x_j, x_{j+1}, x_{j+2}) = x_j^2 + c_{j1}x_{j+1}^2 + c_{j2}x_j + c_{j3}x_{j+1} + c_{j4}x_{j+2} + c_{j0} \equiv 0 \bmod p,$$

where

$$\begin{aligned} c_{j1} &= -L_{j-1}^{-2}L_j^2 \bmod p, \\ c_{j2} &= 2L_{j-1}^{-1}w_{j-1} \bmod p, \end{aligned}$$

$$\begin{aligned}
c_{j3} &= -2L_{j-1}^{-2}L_jw_j - a^{-1}L_{j-1}^{-2}L_j \bmod p, \\
c_{j4} &= a^{-1}L_{j-1}^{-2}L_{j+1} \bmod p, \\
c_{j0} &= L_{j-1}^{-2}(w_{j-1}^2 - w_j^2) - a^{-1}L_{j-1}^{-2}(w_j - w_{j+1}) \bmod p.
\end{aligned}$$

This gives a system of $n - 1$ modular equations in $n + 1$ variables:

$$\begin{cases} f_1(x_1, x_2, x_3) \equiv 0 \bmod p, \\ f_2(x_2, x_3, x_4) \equiv 0 \bmod p, \\ \vdots \\ f_{n-1}(x_{n-1}, x_n, x_{n+1}) \equiv 0 \bmod p. \end{cases}$$

The root of interest is $(x_1^*, \dots, x_{n+1}^*) = (u_0, \dots, u_n)$, with bounds $X_k = p^\delta$ for all k . We apply the Basic Strategy to solve this system.

Theorem 1. *Let a QCG be defined by $F(x) = ax^2 + b$ over $\mathbb{Z}_p$ with known a and unknown b . Given $n + 1$ consecutive truncated outputs for fixed $n \geq 2$, the entire sequence can be predicted, provided that Assumption 1 holds and that the single continuous unknown part of size p^δ for each iteration satisfies*

$$\delta < \frac{n-1}{2n+1} - \varepsilon,$$

for some negligible $\varepsilon > 0$. The time complexity is polynomial in $\log p$. As $n \rightarrow \infty$, the asymptotic bound is $\delta < 1/2$.

Proof. Let m be a large positive integer. Let $\delta_{i,j} = 0$ if $i \neq j$, and $\delta_{i,j} = 1$ if $i = j$. Define the elementary vectors

$$\alpha_j^{(n+1)} = (\delta_{1,j}, \dots, \delta_{n+1,j}) \in \{0, 1\}^{n+1}. \quad (7)$$

The set of monomials for our shift polynomials is determined by the Newton polytopes associated with the system. Let

$$\begin{aligned}
A &= \{\mathbf{0}, \alpha_1^{(n+1)}, \dots, \alpha_{n+1}^{(n+1)}, 2\alpha_1^{(n+1)}, \dots, 2\alpha_n^{(n+1)}\} \subseteq \mathbb{Z}^{n+1}, \\
B &= \{2\alpha_1^{(n+1)}, \dots, 2\alpha_{n-1}^{(n+1)}\} \subseteq \mathbb{Z}^{n+1}.
\end{aligned}$$

Hence, we observe that A has index 1. The set B corresponds to the leading monomials $x_1^2, \dots, x_{n-1}^2$ of the polynomials $f_1, \dots, f_{n-1}$, which will be used in the computation of σ_p .

For each ℓ , we then define the corresponding monomial set

$$S_\ell = \left\{ x_1^{i_1} \cdots x_{n+1}^{i_{n+1}} \mid (i_1, \dots, i_{n+1}) \in (m - \ell)A + \ell B \right\}.$$

We construct a lattice A following the general method described in Section 2. For each monomial $\lambda = x_1^{i_1} \cdots x_{n+1}^{i_{n+1}} \in S_\ell \setminus S_{\ell+1}$, we define the corresponding shift polynomial $g_\lambda(x_1, \dots, x_{n+1})$ as follows.

Step 1. Choose $\ell_1, \dots, \ell_{n-1}$ such that $\sum_{j=1}^{n-1} \ell_j = \ell$ and

$$(i_1, \dots, i_{n+1}) \in (m - \ell)A + \sum_{j=1}^{n-1} \ell_j \alpha_j^{(n+1)}.$$

Step 2. Define $g_\lambda(x_1, \dots, x_{n+1})$ by

$$g_\lambda(x_1, \dots, x_{n+1}) = \frac{\lambda}{\prod_{j=1}^{n-1} \text{LM}(f_j)^{\ell_j}} \cdot \prod_{j=1}^{n-1} f_j^{\ell_j} \cdot p^{m-\ell},$$

The coefficient vectors of $g_\lambda(x_1 X_1, \dots, x_{n+1} X_{n+1})$ are used as the basis of A . The basis matrix of A can be made diagonal, and the solvable condition from (6) becomes

$$\prod_{j=1}^{n+1} X_j^{\sigma_j} p^{\sigma_p} < p^{\sigma_m - \varepsilon_0}, \quad (8)$$

where $\sigma_m, \sigma_j, \sigma_p$ are sums of exponents over all shift polynomials. Let N be the convex hull of A . Using Lemma 3 and Lemma 4, we compute the leading terms of these sums as functions of m .

Computing σ_m . The term σ_m relates to the dimension of the lattice, which is determined by the volume of N by Lemma 3:

$$\lim_{m \rightarrow \infty} \frac{\sigma_m}{m^{n+2}} = \lim_{m \rightarrow \infty} \frac{|mA|}{m^{n+1}} = \int_N 1 \, dV.$$

Let N_0 be the standard $(n+1)$ -dimensional simplex (the convex hull of $\mathbf{0}$ and $\alpha_1^{(n+1)}, \dots, \alpha_{n+1}^{(n+1)}$). Then we have

$$\int_N 1 \, dV = 2^n \cdot \int_{N_0} 1 \, dV.$$

Moreover, the volume of N_0 is

$$\begin{aligned} \int_{N_0} 1 \, dV &= \int_{x_1=0}^1 \int_{x_2=0}^{1-x_1} \int_{x_3=0}^{1-x_1-x_2} \cdots \int_{x_{n+1}=0}^{1-x_1-x_2-\cdots-x_n} 1 \, dx_{n+1} \cdots dx_1 \\ &= \frac{1}{(n+1)!}. \end{aligned}$$

Because N is a scaled version of N_0 along the first n coordinates, the volume of N is 2^n times the volume of N_0 .

$$\lim_{m \rightarrow \infty} \frac{\sigma_m}{m^{n+2}} = \int_N 1 \, dV = 2^n \cdot \int_{N_0} 1 \, dV = \frac{2^n}{(n+1)!}.$$

Therefore, we conclude that

$$\sigma_m = \frac{2^n}{(n+1)!} m^{n+2} + o(m^{n+2}).$$

Computing σ_j . The term σ_j relates to the sum of exponents of x_j over the shift polynomials. The asymptotic relation given by Lemma 4 can be expressed as

$$\lim_{m \rightarrow \infty} \frac{\sigma_j}{m^{n+2}} = \int_N x_j \, dV = \begin{cases} 2^{n+1} \int_{N_0} x_j \, dV & \text{for } j = 1, \dots, n, \\ 2^n \int_{N_0} x_j \, dV & \text{for } j = n+1. \end{cases}$$

Note that by symmetry, each coordinate contributes equally, and hence we have

$$\int_{N_0} x_1 \, dV = \dots = \int_{N_0} x_{n+1} \, dV.$$

It follows that

$$\int_{N_0} x_j \, dV = \frac{1}{n+1} \int_{N_0} \left(\sum_{k=1}^{n+1} x_k \right) dV, \quad \text{for } j = 1, \dots, n+1.$$

Recall that the standard n -simplex is defined as

$$\Delta^n = \left\{ (x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1} \mid x_i \geq 0, \sum_{i=1}^{n+1} x_i = 1 \right\},$$

whose volume is $\int_{\Delta^n} 1 \, dV = 1/n!$. Using this, we compute

$$\int_{N_0} \left(\sum x_k \right) dV = \int_0^1 s \cdot \text{Vol}(s\Delta^n) \, ds = \int_0^1 s \cdot \frac{s^n}{n!} \, ds = \frac{1}{(n+2) \cdot n!}.$$

Therefore, we have

$$\int_{N_0} x_j \, dV = \frac{1}{n+1} \cdot \frac{1}{(n+2) \cdot n!} = \frac{1}{(n+2)!}.$$

Combining the above results, we conclude that

$$\sigma_1 = \sigma_2 = \dots = \sigma_n = 2\sigma_{n+1} = \frac{2^{n+1}}{(n+2)!} m^{n+2} + o(m^{n+2}).$$

Computing σ_p . The term σ_p relates to the sum of powers of p in the lattice construction. This can be computed as the difference between the volumes of two related polytopes. Following the method in [11], we first define $\tilde{A}$ as

$$\begin{aligned} \tilde{A} = \{ & (i_1, \dots, i_{n+1}, 1) \mid (i_1, \dots, i_{n+1}) \in A \} \\ & \cup \{ (i_1, \dots, i_{n+1}, 0) \mid (i_1, \dots, i_{n+1}) \in B \} \subseteq \mathbb{Z}^{n+2}, \end{aligned}$$

and let $\tilde{N}$ be the convex hull of $\tilde{A}$. Hence, we have

$$\lim_{m \rightarrow \infty} \frac{\sigma_p}{m^{n+2}} = \int_N 1 \, dV - \int_{\tilde{N}} 1 \, dV.$$

Thus, we turn to evaluate the integral $\int_{\tilde{N}} 1 \, dV$. As before, we introduce a simpler polytope $\tilde{N}_0$, defined as the convex hull of the following point set

$$\begin{aligned} \tilde{A}_0 = & \{(i_1, \dots, i_{n+1}, 1) \mid (i_1, \dots, i_{n+1}) \in \{\mathbf{0}, \alpha_1^{(n+1)}, \dots, \alpha_{n+1}^{(n+1)}\}\} \\ & \cup \{(i_1, \dots, i_{n+1}, 0) \mid (i_1, \dots, i_{n+1}) \in \{\alpha_1^{(n+1)}, \dots, \alpha_{n-1}^{(n+1)}\}\} \subseteq \mathbb{Z}^{n+2}. \end{aligned}$$

Accordingly, we have

$$\int_{\tilde{N}} 1 \, dV = 2^n \cdot \int_{\tilde{N}_0} 1 \, dV.$$

However, directly computing the integral over $\tilde{N}_0$ remains a challenge. To overcome this, we consider a larger, more symmetric polytope S , defined as the convex hull of $\tilde{A}_0 \cup \{\mathbf{0}\} \subseteq \mathbb{Z}^{n+2}$.

A key observation is that if we denote by T the polytope generated by the points

$$\{\mathbf{0}, \alpha_1^{(n+2)}, \dots, \alpha_{n-1}^{(n+2)}, \alpha_{n+2}^{(n+2)}, (0, \dots, 0, 1, 0, 1), (0, \dots, 0, 1, 1)\} \subseteq \mathbb{Z}^{n+2},$$

that is

$$\{\mathbf{0}, (1, 0, \dots, 0), \dots, (0, \dots, 1, 0, 0, 0), (0, \dots, 0, 1), (0, \dots, 0, 1, 0, 1), (0, \dots, 0, 1, 1)\}.$$

Then it follows that

$$\int_{\tilde{N}_0} 1 \, dV = \int_S 1 \, dV - \int_T 1 \, dV.$$

The volume of T is straightforward to compute, and it is

$$\int_T 1 \, dV = \frac{1}{(n+2)!}.$$

For the computation of $\int_S 1 \, dV$, we provide further details. Note that the projection of S onto the first $n-1$ coordinates is a polytope generated by

$$\{\mathbf{0}, \alpha_1^{(n-1)}, \dots, \alpha_{n-1}^{(n-1)}\} \subseteq \mathbb{Z}^{n-1},$$

whose volume is straightforward to compute, namely $1/(n-1)!$.

Another key observation is that if we consider the projection of S onto the first $n-1$ coordinates along with the $(n+2)$ -th coordinate, i.e., define the projection map ψ as

$$\begin{aligned} \psi : \mathbb{Z}^{n+2} &\longrightarrow \mathbb{Z}^n, \\ (i_1, \dots, i_{n+2}) &\longmapsto (i_1, \dots, i_{n-1}, i_{n+2}), \end{aligned}$$

$\psi(S)$ forms an n -dimensional prism of height 1 with base that is the polytope generated by

$$\{\mathbf{0}, \alpha_1^{(n-1)}, \dots, \alpha_{n-1}^{(n-1)}\} \subseteq \mathbb{Z}^{n-1}.$$

Thus, the volume of $\psi(S)$ remains $1/(n-1)!$ and we have

$$\int_S 1 \, dV = \int_{x_{n+1}=0}^1 \int_{x_n=0}^{1-x_{n+1}} \frac{(1-x_n-x_{n+1})^n}{(n-1)!} \, dx_n \, dx_{n+1} = \frac{n}{(n+2)!}.$$

It follows that

$$\begin{aligned} \lim_{m \rightarrow \infty} \frac{\sigma_p}{m^{n+2}} &= \int_N 1 \, dV - \int_{\tilde{N}} 1 \, dV = \int_N 1 \, dV - 2^n \cdot \left(\int_S 1 \, dV - \int_T 1 \, dV \right) \\ &= \frac{2^n}{(n+1)!} - \frac{2^n \cdot (n-1)}{(n+2)!} \\ &= \frac{3 \cdot 2^n}{(n+2)!}. \end{aligned}$$

Therefore, we conclude that

$$\sigma_p = \frac{3 \cdot 2^n}{(n+2)!} m^{n+2} + o(m^{n+2}).$$

Plugging these values into the solvable condition (8) (with $X_j = p^\delta$), taking the logarithm with base p , and keeping only the leading terms in m , we get

$$\delta \left(n \cdot \frac{2^{n+1}}{(n+2)!} + \frac{2^n}{(n+2)!} \right) + \frac{3 \cdot 2^n}{(n+2)!} < \frac{2^n}{(n+1)!} - \varepsilon_1.$$

Multiplying by $(n+2)!$ and dividing by 2^n simplifies this to

$$\delta(2n+1) + 3 < (n+2) - \varepsilon_2,$$

which gives the final bound

$$\delta < \frac{n-1}{2n+1} - \varepsilon.$$

The overall time complexity is dominated by both the LLL reduction and root extraction. For fixed m and n , it is polynomial in $\log p$. $\square$

Experimental Validation. We demonstrate the validity of our attack with experiments for $\pi = \log_2(p) = 256$. The results are shown in Table 2.

Table 2. Experimental results of attacks on QCGs with known a and unknown b

π (bits)	n	κ (bits)	δ (expt.)	m	ω	Time (seconds)
256	2	203	0.207	2	19	0.322
256	3	190	0.257	2	33	0.671
256	4	184	0.281	2	51	1.807

3.2 The Case of Unknown a and Known b

We now consider the case of unknown a and known b in the QCG defined by $F(x) = ax^2 + b$ over $\mathbb{Z}_p$. We start with the system of equations $v_j = av_{j-1}^2 + b \bmod p$ for $j = 1, \dots, n$. We can eliminate the unknown a by taking ratios: for $j = 1, \dots, n-1$, we have

$$\frac{v_j - b}{v_{j+1} - b} = \frac{v_{j-1}^2}{v_j^2} \bmod p,$$

which can be rearranged to $v_j^2(v_j - b) - v_{j-1}^2(v_{j+1} - b) \equiv 0 \bmod p$. Substituting $v_k = w_k + L_k u_k$ leads to

$$(w_j + L_j u_j)^2(w_j + L_j u_j - b) - (w_{j-1} + L_{j-1} u_{j-1})^2(w_{j+1} + L_{j+1} u_{j+1} - b) \equiv 0 \bmod p.$$

Letting $x_k = u_{k-1}$ for $k = 1, \dots, n+1$, we obtain for each $j = 1, \dots, n-1$ a modular polynomial equation $f_j(x_j, x_{j+1}, x_{j+2}) \equiv 0 \bmod p$. After making the equation monic in x_{j+1} by multiplying by $L_j^{-3} \bmod p$, the leading monomial is x_{j+1}^3 . The polynomial is

$$\begin{aligned} f_j(x_j, x_{j+1}, x_{j+2}) = & x_{j+1}^3 + c_{j1}x_j^2x_{j+2} + c_{j2}x_j^2 + c_{j3}x_{j+1}^2 + c_{j4}x_jx_{j+2} \\ & + c_{j5}x_j + c_{j6}x_{j+1} + c_{j7}x_{j+2} + c_{j0} \equiv 0 \bmod p, \end{aligned}$$

where the coefficients c_{jk} are known. This gives a system of $n-1$ modular equations in $n+1$ variables.

$$\begin{cases} f_1(x_1, x_2, x_3) \equiv 0 \bmod p, \\ f_2(x_2, x_3, x_4) \equiv 0 \bmod p, \\ \vdots \\ f_{n-1}(x_{n-1}, x_n, x_{n+1}) \equiv 0 \bmod p. \end{cases}$$

The root of interest is $(x_1^*, \dots, x_{n+1}^*) = (u_0, \dots, u_n)$, with bounds $X_k = p^\delta$ for all k . We apply the Basic Strategy. The analysis is analogous to that in Section 3.1, but with different Newton polytopes reflecting the cubic leading terms.

Theorem 2. *Let a QCG be defined by $F(x) = ax^2 + b$ over $\mathbb{Z}_p$ with unknown a and known b . Given $n+1$ consecutive truncated outputs for fixed $n \geq 2$, the entire sequence can be predicted, provided that Assumption 1 holds and that the single continuous unknown part of size p^δ for each iteration satisfies*

$$\delta < \frac{n-1}{3n+3} - \varepsilon,$$

for some negligible ε . The time complexity is polynomial in $\log p$. As $n \rightarrow \infty$, the asymptotic bound is $\delta < 1/3$.

Proof. The proof structure follows that of Theorem 1. The leading monomial of each f_j in the graded reverse lexicographic order is x_{j+1}^3 . Let m be a large positive integer, and define the same elementary vectors $\alpha_j^{(n+1)} \in \mathbb{Z}^{n+1}$ as (7). The sets A and B are defined as

$$\begin{aligned} A &= \{\mathbf{0}, \alpha_1^{(n+1)}, \dots, \alpha_{n+1}^{(n+1)}, 3\alpha_1^{(n+1)}, \dots, 3\alpha_{n+1}^{(n+1)}\} \subseteq \mathbb{Z}^{n+1}, \\ B &= \{3\alpha_2^{(n+1)}, \dots, 3\alpha_n^{(n+1)}\} \subseteq \mathbb{Z}^{n+1}. \end{aligned}$$

Note that we include $3\alpha_1^{(n+1)}$ and $3\alpha_{n+1}^{(n+1)}$ in A to simplify the volume calculations, as the polynomials f_j have degree 3 in all variables. Similar to the proof of Theorem 1, we use A and B to construct the shift polynomials $g_\lambda(x_1, \dots, x_{n+1})$ for each monomial λ , and form the lattice Λ using the coefficient vectors of $g_\lambda(x_1 X_1, \dots, x_{n+1} X_{n+1})$ as its basis. Observe that the basis matrix of Λ is diagonal. Hence, the solvable condition from (6) becomes

$$\prod_{j=1}^{n+1} X_j^{\sigma_j} p^{\sigma_p} < p^{\sigma_m - \varepsilon_0}.$$

It remains to compute the values of σ_m , $\sigma_1, \dots, \sigma_{n+1}$, and σ_p .

Computing σ_m . To evaluate σ_m , we denote by N the convex hull of A and apply the asymptotic estimate from Lemma 3:

$$\lim_{m \rightarrow \infty} \frac{\sigma_m}{m^{n+2}} = \lim_{m \rightarrow \infty} \frac{|mA|}{m^{n+1}} = \int_N 1 \, dV.$$

Let N_0 denote the convex hull of the points $\{\mathbf{0}, \alpha_1^{(n+1)}, \dots, \alpha_{n+1}^{(n+1)}\}$. It follows that

$$\int_N 1 \, dV = 3^{n+1} \cdot \int_{N_0} 1 \, dV = \frac{3^{n+1}}{(n+1)!}.$$

Therefore, we obtain

$$\sigma_m = \frac{3^{n+1}}{(n+1)!} m^{n+2} + o(m^{n+2}).$$

Computing σ_j . For each σ_j , we apply Lemma 4, which gives

$$\lim_{m \rightarrow \infty} \frac{\sigma_j}{m^{n+2}} = \int_N x_j \, dV.$$

We have

$$\int_N x_j \, dV = 3^{n+2} \cdot \int_{N_0} x_j \, dV = \frac{3^{n+2}}{(n+2)!},$$

where N_0 is the convex hull of $\{\mathbf{0}, \alpha_1^{(n+1)}, \dots, \alpha_{n+1}^{(n+1)}\}$ as before. Hence,

$$\sigma_1 = \dots = \sigma_{n+1} = \frac{3^{n+2}}{(n+2)!} m^{n+2} + o(m^{n+2}).$$

Computing σ_p . To compute σ_p , we define $\tilde{A}$ similarly to the proof of Theorem 1 as follows:

$$\begin{aligned} \tilde{A} = \{ & (i_1, \dots, i_{n+1}, 1) \mid (i_1, \dots, i_{n+1}) \in A \} \\ & \cup \{ (i_1, \dots, i_{n+1}, 0) \mid (i_1, \dots, i_{n+1}) \in B \} \subseteq \mathbb{Z}^{n+2}. \end{aligned}$$

Let $\tilde{N}$ be the convex hull of $\tilde{A}$. Then, we have

$$\lim_{m \rightarrow \infty} \frac{\sigma_p}{m^{n+2}} = \int_N 1 \, dV - \int_{\tilde{N}} 1 \, dV.$$

Our task reduces to evaluating $\int_{\tilde{N}} 1 \, dV$. Similarly, we introduce a simplified polytope $\tilde{N}_0$, defined as the convex hull of the following point set

$$\begin{aligned} \tilde{A}_0 = \{ & (i_1, \dots, i_{n+1}, 1) \mid (i_1, \dots, i_{n+1}) \in \{\mathbf{0}, \alpha_1^{(n+1)}, \dots, \alpha_{n+1}^{(n+1)}\} \} \\ & \cup \{ (i_1, \dots, i_{n+1}, 0) \mid (i_1, \dots, i_{n+1}) \in \{\alpha_2^{(n+1)}, \dots, \alpha_n^{(n+1)}\} \} \subseteq \mathbb{Z}^{n+2}. \end{aligned}$$

It implies that

$$\int_{\tilde{N}} 1 \, dV = 3^{n+1} \cdot \int_{\tilde{N}_0} 1 \, dV.$$

Following the same idea used in Theorem 1, let S denote the convex hull of $\tilde{A}_0 \cup \{\mathbf{0}\} \subseteq \mathbb{Z}^{n+2}$, and define T as the simplex generated by the points

$$\{\mathbf{0}, \alpha_2^{(n+2)}, \dots, \alpha_n^{(n+2)}, \alpha_{n+2}^{(n+2)}, (1, 0, \dots, 0, 1), (0, \dots, 0, 1, 1)\} \subseteq \mathbb{Z}^{n+2},$$

which has volume $1/(n+2)!$. Then, we have

$$\int_{\tilde{N}_0} 1 \, dV = \int_S 1 \, dV - \int_T 1 \, dV.$$

For the computation of $\int_S 1 \, dV$, We define the projection map ψ by taking the components from the second to the n -th together with the $(n+2)$ -nd component, that is,

$$\begin{aligned} \psi : \mathbb{Z}^{n+2} &\longrightarrow \mathbb{Z}^n, \\ (i_1, \dots, i_{n+2}) &\longmapsto (i_2, \dots, i_n, i_{n+2}), \end{aligned}$$

$\psi(S)$ forms an $n-1$ -dimensional prism of height 1 with base that is the polytope generated by

$$\{\mathbf{0}, \alpha_1^{(n-1)}, \dots, \alpha_{n-1}^{(n-1)}\} \subseteq \mathbb{Z}^{n-1}.$$

Thus, the volume of $\psi(S)$ remains $1/(n-1)!$ and we have

$$\int_S 1 \, dV = \int_{x_{n+1}=0}^1 \int_{x_1=0}^{1-x_{n+1}} \frac{(1-x_1-x_{n+1})^{n-1}}{(n-2)!} \, dx_1 \, dx_{n+1} = \frac{n}{(n+2)!}.$$

Combining the above results, we conclude that

$$\sigma_p = \frac{3^{n+2}}{(n+2)!} m^{n+2} + o(m^{n+2}).$$

Plugging these into the solvable condition gives

$$\delta \cdot (n+1) \cdot \frac{3^{n+2}}{(n+2)!} + \frac{3^{n+2}}{(n+2)!} < \frac{3^{n+1}}{(n+1)!} - \varepsilon_1.$$

Multiplying by $(n+2)!$ and dividing by 3^{n+1} simplifies this to

$$\delta \cdot 3(n+1) + 3 < (n+2) - \varepsilon_2,$$

which yields the bound

$$\delta < \frac{n-1}{3n+3} - \varepsilon.$$

The complexity analysis is similar to that of Theorem 1. Therefore, the time complexity is polynomial in $\log p$. $\square$

Remark 2. In contrast to Theorem 1, where the monomials have degree at most 2, the polynomials here involve cubic terms. This requires defining the set A using points scaled by a factor of 3 and adding additional points to ensure that the resulting lattice remains diagonal.

Experimental Validation. We demonstrate the validity of our attack with experiments for $\pi = \log_2(p) = 256$. The results are shown in Table 3.

Table 3. Experimental results of attacks on QCGs with unknown a and known b

π (bits)	n	κ (bits)	δ (expt.)	m	ω	Time (seconds)
256	2	234	0.086	2	36	0.533
256	3	229	0.105	2	82	3.695
256	4	226	0.117	2	148	15.143

4 Attacks on Power Generators

Let p be a prime number of bit-length π , and let $\mathbb{Z}_p$ denote the ring of integers modulo p . A Power Generator is defined by the recurrence $v_{i+1} = F(v_i) \bmod p$, where $F(x) = x^e$ is a polynomial of a fixed public exponent e over $\mathbb{Z}_p[x]$, and v_0 is the secret seed. We assume that the generator reveals the κ truncated output bits of each state v_i per iteration for $\kappa \in \{1, \dots, \pi\}$ with known output pattern. This implies that the internal state can be written as

$$v_i = w_i + L_i u_i,$$

where w_i is the known partial output, L_i is a power of 2 denoting the output LSB block, and $u_i \leq 2^{\pi-k} - 1 = p^\delta$ is the unknown portion. Our goal is to recover u_i for given infinitely many consecutive outputs, and to maximize the admissible value of δ .

We consider an attack using $n+1$ consecutive outputs. This gives n relations $v_j = v_{j-1}^e \bmod p$ for $j = 1, \dots, n$. As in the previous sections, we must eliminate unknown constants (if any) and relate consecutive states. For the standard Power Generator, no elimination is needed and we directly obtain $v_{j-1}^e - v_j = 0 \bmod p$. This leads to a system of n modular equations. After making the polynomial monic in x_j , the leading monomial is x_j^e . Specifically, each modular equation is of the form

$$f_j(x_j, x_{j+1}) = x_j^e + c_{i,1}x_i^{e-1} + \dots + c_{i,e-1}x_i + c_{i,e}x_{i+1} + c_{i,0} \equiv 0 \bmod p.$$

Hence, the system of n modular equations to be solved is

$$\begin{cases} f_1(x_1, x_2) \equiv 0 \bmod p, \\ f_2(x_2, x_3) \equiv 0 \bmod p, \\ \vdots \\ f_n(x_n, x_{n+1}) \equiv 0 \bmod p. \end{cases}$$

The root of interest is $(x_1^*, \dots, x_{n+1}^*) = (u_0, \dots, u_n)$, with bounds $X_k = p^\delta$ for all k . We apply the Basic Strategy to solve this system.

Theorem 3. *Let a Power Generator be defined by $F(x) = x^e$ over $\mathbb{Z}_p$ with a fixed public e . Given $n+1$ consecutive truncated outputs for fixed $n \geq 1$, the entire sequence can be predicted, provided that Assumption 1 holds and that the single continuous unknown part of size p^δ for each iteration satisfies*

$$\delta < \frac{n}{en+1} - \varepsilon,$$

for some negligible ε . The time complexity is polynomial in $\log p$. As $n \rightarrow \infty$, the asymptotic bound is $\delta < 1/e$.

Proof. The proof is analogous to that of Theorem 1. The leading monomial of each derived polynomial f_j is x_j^e . Let m be a large positive integer, and define the same elementary vectors $\alpha_j^{(n+1)} \in \mathbb{Z}^{n+1}$ as (7). The sets A and B are defined as

$$\begin{aligned} A &= \{\mathbf{0}, \alpha_1^{(n+1)}, \dots, \alpha_{n+1}^{(n+1)}, e\alpha_1^{(n+1)}, \dots, e\alpha_n^{(n+1)}\} \subseteq \mathbb{Z}^{n+1}, \\ B &= \{e\alpha_1^{(n+1)}, \dots, e\alpha_n^{(n+1)}\} \subseteq \mathbb{Z}^{n+1}. \end{aligned}$$

Similar to the proof of Theorem 1, we use A and B to construct the shift polynomials $g_\lambda(x_1, \dots, x_{n+1})$ for each monomial λ , and form the lattice Λ using

the coefficient vectors of $g_\lambda(x_1 X_1, \dots, x_{n+1} X_{n+1})$ as its basis. Observe that the basis matrix of A is diagonal. Hence, the solvable condition from (6) becomes

$$\prod_{j=1}^{n+1} X_j^{\sigma_j} p^{\sigma_p} < p^{\sigma_m - \varepsilon_0}.$$

It remains to compute the values of $\sigma_m, \sigma_1, \dots, \sigma_{n+1}$, and σ_p .

Computing σ_m . To evaluate σ_m , we denote by N the convex hull of A and apply the asymptotic estimate from Lemma 3:

$$\lim_{m \rightarrow \infty} \frac{\sigma_m}{m^{n+2}} = \lim_{m \rightarrow \infty} \frac{|mA|}{m^{n+1}} = \int_N 1 \, dV.$$

Let N_0 denote the convex hull of the points $\{\mathbf{0}, \alpha_1^{(n+1)}, \dots, \alpha_{n+1}^{(n+1)}\}$. It follows that

$$\int_N 1 \, dV = e^n \cdot \int_{N_0} 1 \, dV = \frac{e^n}{(n+1)!}.$$

Therefore, we obtain

$$\sigma_m = \frac{e^n}{(n+1)!} m^{n+2} + o(m^{n+2}).$$

Computing σ_j . The asymptotic relation given by Lemma 4 can be expressed as

$$\lim_{m \rightarrow \infty} \frac{\sigma_j}{m^{n+2}} = \int_N x_j \, dV = \begin{cases} e^{n+1} \int_{N_0} x_j \, dV & \text{for } j = 1, \dots, n, \\ e^n \int_{N_0} x_j \, dV & \text{for } j = n+1. \end{cases}$$

Note that by symmetry, each coordinate contributes equally, and hence we have

$$\int_{N_0} x_1 \, dV = \dots = \int_{N_0} x_{n+1} \, dV.$$

It follows that

$$\int_{N_0} x_j \, dV = \frac{1}{n+1} \int_{N_0} \left(\sum_{k=1}^{n+1} x_k \right) dV, \quad \text{for } j = 1, \dots, n+1.$$

Similarly, we conclude that

$$\sigma_1 = \sigma_2 = \dots = \sigma_n = e \sigma_{n+1} = \frac{e^{n+1}}{(n+2)!} m^{n+2} + o(m^{n+2}).$$

Computing σ_p . This can be computed as the difference between the volumes of two related polytopes. Following the method in [11], and similar to the proof of Theorem 1, we first define $\tilde{A}$ as

$$\begin{aligned} \tilde{A} = \{ & (i_1, \dots, i_{n+1}, 1) \mid (i_1, \dots, i_{n+1}) \in A \} \\ & \cup \{ (i_1, \dots, i_{n+1}, 0) \mid (i_1, \dots, i_{n+1}) \in B \} \subseteq \mathbb{Z}^{n+2}, \end{aligned}$$

and let $\tilde{N}$ be the convex hull of $\tilde{A}$. Hence, we have

$$\lim_{m \rightarrow \infty} \frac{\sigma_p}{m^{n+2}} = \int_N 1 \, dV - \int_{\tilde{N}} 1 \, dV.$$

Thus, we turn to evaluate the integral $\int_{\tilde{N}} 1 \, dV$. As before, we introduce a simpler polytope $\tilde{N}_0$, defined as the convex hull of the following point set

$$\begin{aligned} \tilde{A}_0 = \{ & (i_1, \dots, i_{n+1}, 1) \mid (i_1, \dots, i_{n+1}) \in \{\mathbf{0}, \alpha_1^{(n+1)}, \dots, \alpha_{n+1}^{(n+1)}\} \} \\ & \cup \{ (i_1, \dots, i_{n+1}, 0) \mid (i_1, \dots, i_{n+1}) \in \{\alpha_1^{(n+1)}, \dots, \alpha_n^{(n+1)}\} \} \subseteq \mathbb{Z}^{n+2}. \end{aligned}$$

Accordingly, we have

$$\int_{\tilde{N}} 1 \, dV = e^n \cdot \int_{\tilde{N}_0} 1 \, dV.$$

Following the same technique used in the proof of Theorem 1, let S denote the convex hull of $\tilde{A}_0 \cup \{\mathbf{0}\} \subseteq \mathbb{Z}^{n+2}$, and define T as the simplex generated by the points,

$$\{\mathbf{0}, \alpha_1^{(n+2)}, \dots, \alpha_n^{(n+2)}, \alpha_{n+2}^{(n+2)}, (0, \dots, 1, 1)\} \subseteq \mathbb{Z}^{n+2}.$$

The volume of T is $\int_T 1 \, dV = 1/(n+2)!$ as shown before, and we have

$$\int_{\tilde{N}_0} 1 \, dV = \int_S 1 \, dV - \int_T 1 \, dV.$$

For the computation of $\int_S 1 \, dV$, we consider the projection of S onto the first n coordinates along with the $(n+2)$ -th coordinate, i.e., define the projection map ψ as

$$\begin{aligned} \psi : \mathbb{Z}^{n+2} & \longrightarrow \mathbb{Z}^{n+1}, \\ (i_1, \dots, i_{n+2}) & \longmapsto (i_1, \dots, i_n, i_{n+2}), \end{aligned}$$

$\psi(S)$ forms an n -dimensional prism of height 1 with base that is the polytope generated by

$$\{\mathbf{0}, \alpha_1^{(n)}, \dots, \alpha_n^{(n)}\} \subseteq \mathbb{Z}^n.$$

Thus, the volume of $\psi(S)$ remains $1/n!$ and we have

$$\int_S 1 \, dV = \int_{x_{n+1}=0}^1 \frac{(1-x_{n+1})^{n+1}}{n!} \, dx_{n+1} = \frac{n+1}{(n+2)!}.$$

Combining the above results, we conclude that

$$\sigma_p = \frac{2 \cdot e^n}{(n+2)!} m^{n+2} + o(m^{n+2}).$$

Plugging these into the solvable condition with $X_j = p^\delta$ gives

$$\delta \left(n \cdot \frac{e^{n+1}}{(n+2)!} + \frac{e^n}{(n+2)!} \right) + \frac{2e^n}{(n+2)!} < \frac{e^n}{(n+1)!} - \varepsilon_1,$$

which simplifies to

$$\delta(en+1) + 2 < (n+2) - \varepsilon_2,$$

leading to the bound

$$\delta < \frac{n}{en+1} - \varepsilon.$$

The time complexity is polynomial in $\log p$. □

Experimental Validation. We demonstrate the validity of our attack with experiments for $\pi = \log_2(p) = 256$. The results are shown in Table 4.

Table 4. Experimental results of attacks on standard Power Generators

π (bits)	e	n	κ (bits)	δ (expt.)	m	ω	Time (seconds)
256	3	1	208	0.187	3	22	0.382
256	3	2	204	0.203	3	76	1.951
256	3	3	202	0.210	3	193	9.851

We further show that the same bound of $1/e$ can be achieved even when the Power Generator map polynomial $F(x) = x^e$ is perturbed by an unknown masking constant c . We consider a perturbed Power Generator defined by $v_j = v_{j-1}^e + c \bmod p$ for $j = 1, \dots, n$, where c is unknown. By subtracting consecutive relations, we eliminate c : $v_j - v_{j+1} = v_{j-1}^e - v_j^e \bmod p$. This can be rewritten as

$$v_{j-1}^e - v_j^e - v_j + v_{j+1} \equiv 0 \bmod p.$$

We substitute $v_k = w_k + L_k u_k$ and let $x_k = u_{k-1}$ to get a system of $n-1$ modular equations. After making each polynomial $f_j(x_j, x_{j+1}, x_{j+2})$ monic in x_j by multiplying by $L_{j-1}^{-e} \bmod p$, the leading monomial is x_j^e . Hence, we have

$$\begin{aligned} f_j(x_j, x_{j+1}, x_{j+2}) &= x_j^e + c_{j,1} x_j^{e-1} + \dots + c_{j,e-1} x_j \\ &+ c_{j,e} x_{j+1}^e + \dots + c_{j,2e-1} x_{j+1} + c_{j,2e} x_{j+2} + c_{j,0} \equiv 0 \bmod p. \end{aligned}$$

The coefficients $c_{j,k}$ can be easily figured out. Therefore, we aim to solve the following modular equation system,

$$\begin{cases} f_1(x_1, x_2, x_3) \equiv 0 \pmod{p}, \\ f_2(x_2, x_3, x_4) \equiv 0 \pmod{p}, \\ \vdots \\ f_{n-1}(x_{n-1}, x_n, x_{n+1}) \equiv 0 \pmod{p}. \end{cases}$$

The corresponding root is $(x_1^*, x_2^*, \dots, x_{n+1}^*) = (u_0, u_1, \dots, u_n)$ and the respective upper bounds are set as $X_j = p^\delta$. Similarly, we apply the Basic Strategy along with the asymptotic computing tool. The attack statement is concluded as follows.

Theorem 4. *Let a perturbed Power Generator be defined by $F(x) = x^e + c$ over $\mathbb{Z}_p$ with unknown c . Given $n + 1$ consecutive truncated outputs for fixed $n \geq 2$, the entire sequence can be predicted, provided that Assumption 1 holds and that the single continuous unknown part of size p^δ for each iteration satisfies*

$$\delta < \frac{n-1}{en+1} - \varepsilon,$$

for some negligible $\varepsilon > 0$. The time complexity is polynomial in $\log p$. As $n \rightarrow \infty$, the asymptotic bound is $\delta < 1/e$.

Proof. The proof is analogous to that of Theorem 1. The leading monomial of each derived polynomial f_j is x_j^e . Let m be a large positive integer, and define the same elementary vectors $\alpha_j^{(n+1)} \in \mathbb{Z}^{n+1}$ as (7). The sets A and B are defined as

$$\begin{aligned} A &= \{\mathbf{0}, \alpha_1^{(n+1)}, \dots, \alpha_{n+1}^{(n+1)}, e\alpha_1^{(n+1)}, \dots, e\alpha_n^{(n+1)}\} \subseteq \mathbb{Z}^{n+1}, \\ B &= \{e\alpha_1^{(n+1)}, \dots, e\alpha_{n-1}^{(n+1)}\} \subseteq \mathbb{Z}^{n+1}. \end{aligned}$$

Similar to the proof of Theorem 1, we use A and B to construct the shift polynomials $g_\lambda(x_1, \dots, x_{n+1})$ for each monomial λ , and form the lattice Λ using the coefficient vectors of $g_\lambda(x_1 X_1, \dots, x_{n+1} X_{n+1})$ as its basis. Observe that the basis matrix of Λ is diagonal. Hence, the solvable condition from (6) becomes

$$\prod_{j=1}^{n+1} X_j^{\sigma_j} p^{\sigma_p} < p^{\sigma_m - \varepsilon_0}.$$

It remains to compute the values of $\sigma_m, \sigma_1, \dots, \sigma_{n+1}$, and σ_p .

Computing σ_m . To evaluate σ_m , we denote by N the convex hull of A and apply the asymptotic estimate from Lemma 3:

$$\lim_{m \rightarrow \infty} \frac{\sigma_m}{m^{n+2}} = \lim_{m \rightarrow \infty} \frac{|mA|}{m^{n+1}} = \int_N 1 \, dV.$$

Let N_0 denote the convex hull of the points $\{\mathbf{0}, \alpha_1^{(n+1)}, \dots, \alpha_{n+1}^{(n+1)}\}$. It follows that

$$\int_N 1 \, dV = e^n \cdot \int_{N_0} 1 \, dV = \frac{e^n}{(n+1)!}.$$

Therefore, we obtain

$$\sigma_m = \frac{e^n}{(n+1)!} m^{n+2} + o(m^{n+2}).$$

Computing σ_j . The asymptotic relation given by Lemma 4 can be expressed as

$$\lim_{m \rightarrow \infty} \frac{\sigma_j}{m^{n+2}} = \int_N x_j \, dV = \begin{cases} e^{n+1} \int_{N_0} x_j \, dV & \text{for } j = 1, \dots, n, \\ e^n \int_{N_0} x_j \, dV & \text{for } j = n+1. \end{cases}$$

Similarly, we conclude that

$$\sigma_1 = \sigma_2 = \dots = \sigma_n = e \sigma_{n+1} = \frac{e^{n+1}}{(n+2)!} m^{n+2} + o(m^{n+2}).$$

Computing σ_p . This can be computed as the difference between the volumes of two related polytopes. Following the method in [11], and similar to the proof of Theorem 1, we first define $\tilde{A}$ as

$$\begin{aligned} \tilde{A} = \{ & (i_1, \dots, i_{n+1}, 1) \mid (i_1, \dots, i_{n+1}) \in A \} \\ & \cup \{ (i_1, \dots, i_{n+1}, 0) \mid (i_1, \dots, i_{n+1}) \in B \} \subseteq \mathbb{Z}^{n+2}, \end{aligned}$$

and let $\tilde{N}$ be the convex hull of $\tilde{A}$. Hence, we have

$$\lim_{m \rightarrow \infty} \frac{\sigma_p}{m^{n+2}} = \int_N 1 \, dV - \int_{\tilde{N}} 1 \, dV.$$

Thus, we turn to evaluate the integral $\int_{\tilde{N}} 1 \, dV$. As before, we introduce a simpler polytope $\tilde{N}_0$, defined as the convex hull of the following point set

$$\begin{aligned} \tilde{A}_0 = \{ & (i_1, \dots, i_{n+1}, 1) \mid (i_1, \dots, i_{n+1}) \in \{\mathbf{0}, \alpha_1^{(n+1)}, \dots, \alpha_{n+1}^{(n+1)}\} \} \\ & \cup \{ (i_1, \dots, i_{n+1}, 0) \mid (i_1, \dots, i_{n+1}) \in \{\alpha_1^{(n+1)}, \dots, \alpha_{n-1}^{(n+1)}\} \} \subseteq \mathbb{Z}^{n+2}. \end{aligned}$$

Accordingly, we have

$$\int_{\tilde{N}} 1 \, dV = e^n \cdot \int_{\tilde{N}_0} 1 \, dV.$$

Note that $\tilde{N}_0$ is the same as that used in the proof of Theorem 1. Therefore, we conclude that

$$\sigma_p = \frac{3 \cdot e^n}{(n+2)!} m^{n+2} + o(m^{n+2}).$$

Plugging these into the solvable condition with $X_j = p^\delta$ gives

$$\delta \left(n \cdot \frac{e^{n+1}}{(n+2)!} + \frac{e^n}{(n+2)!} \right) + \frac{3e^n}{(n+2)!} < \frac{e^n}{(n+1)!} - \varepsilon_1,$$

which simplifies to

$$\delta(en + 1) + 3 < (n + 2) - \varepsilon_2,$$

leading to the bound

$$\delta < \frac{n - 1}{en + 1} - \varepsilon.$$

The time complexity is polynomial in $\log p$. □

Experimental Validation. We demonstrate the validity of our attack with experiments for $\pi = \log_2(p) = 256$. The results are shown in Table 5.

Table 5. Experimental results of attacks on perturbed Power Generators

π (bits)	e	n	κ (bits)	δ (expt.)	m	ω	Time (seconds)
256	3	2	230	0.101	2	30	0.423
256	3	3	224	0.125	2	57	1.873
256	3	4	221	0.136	2	93	5.284

5 Attacks on Pollard Generators

A Pollard Generator is defined by the recurrence $v_{i+1} = v_i^2 + c \bmod p$. We consider the case where the parameter c is unknown. Given $n + 1$ consecutive outputs, we have n relations $v_j = v_{j-1}^2 + c \bmod p$. We can eliminate c by subtracting consecutive relations: $v_j - v_{j+1} = v_{j-1}^2 - v_j^2 \bmod p$, which can be rearranged to

$$v_{j-1}^2 - v_j^2 - v_j + v_{j+1} \equiv 0 \bmod p.$$

This is the same relation as that derived for QCGs with known $a = 1$ and unknown b . We substitute $v_k = w_k + L_k u_k$ and let $x_k = u_{k-1}$ to obtain a system of $n - 1$ modular equations of the form

$$f_j(x_j, x_{j+1}, x_{j+2}) = x_j^2 + c_{j1}x_j + c_{j2}x_{j+1}^2 + c_{j3}x_{j+1} + c_{j4}x_{j+2} + c_{j0} \equiv 0 \bmod p.$$

This system is identical in structure to the one in Section 3.1. Therefore, the same analysis applies.

Theorem 5. *Let a Pollard generator be defined by $F(x) = x^2 + c$ over $\mathbb{Z}_p$ with unknown c . Given $n + 1$ consecutive truncated outputs for fixed $n \geq 2$, the entire*

sequence can be predicted, provided that Assumption 1 holds and that the single continuous unknown part of size p^δ for each iteration satisfies

$$\delta < \frac{n-1}{2n+1} - \varepsilon,$$

for some negligible $\varepsilon > 0$. The time complexity is polynomial in $\log p$. As $n \rightarrow \infty$, the asymptotic bound is $\delta < 1/2$.

Proof. The derived polynomial system is a special case of the system for QCGs with known $a = 1$ and unknown b . The proof is therefore identical to that of Theorem 1. $\square$

Experimental Validation. We demonstrate the validity of our attack with experiments for $\pi = \log_2(p) = 256$. The results are shown in Table 6.

Table 6. Experimental results of attacks on Pollard Generators with unknown c

π (bits)	n	κ (bits)	δ (expt.)	m	ω	Time (seconds)
256	2	202	0.210	3	44	0.926
256	3	187	0.269	3	96	5.564
256	4	182	0.289	3	180	44.109

6 Concluding Remarks

We present new asymptotic attack bounds against several common non-linear polynomial congruential generators (PCGs), including QCGs, Power Generators, and Pollard Generators. The first analyses for several attack scenarios with partially known coefficients are provided, improving the overall understanding of their security thresholds. Moreover, our findings are supported by numerical experiments, which confirm the validity and efficacy of these attacks.

We elaborate a new method for applying automated Coppersmith techniques to symbolic systems of equations. Specifically, we target the recursive algebraic structures that emerge from analyzing $n + 1$ consecutive generator outputs. By developing new techniques to symbolically compute the associated high-dimensional polytopes' volumes concerning n , we derive general expressions for the attack bounds, rather than just numerical estimates for fixed systems.

This symbolic analysis framework for structured polynomial systems may be of independent interest. We hope it can find broader applications in other areas of public-key cryptanalysis where similar algebraic structures appear.

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A Proofs for Existing Asymptotic Bounds

A.1 QCG with Known a, b

A QCG is defined by the recurrence $v_{i+1} = av_i^2 + b \bmod p$. We consider the case where the parameters a, b are known. Given $n + 1$ consecutive outputs, we have n relations $v_j = av_{j-1}^2 + b \bmod p$. There is no need to eliminate any coefficients, and we directly get the consecutive relations:

$$av_{j-1}^2 - v_j + b \equiv 0 \bmod p.$$

We substitute $v_k = w_k + L_k u_k$ and let $x_k = u_{k-1}$ to obtain a monic modular equations of the form

$$f_j(x_j, x_{j+1}) = x_j^2 + c_{j1}x_j + c_{j2}x_{j+1} + c_{j0} \equiv 0 \bmod p.$$

Hence, the system of n modular equations to be solved is

$$\begin{cases} f_1(x_1, x_2) \equiv 0 \pmod{p}, \\ f_2(x_2, x_3) \equiv 0 \pmod{p}, \\ \vdots \\ f_n(x_n, x_{n+1}) \equiv 0 \pmod{p}. \end{cases}$$

The root of interest is $(x_1^*, \dots, x_{n+1}^*) = (u_0, \dots, u_n)$, with bounds $X_k = p^\delta$ for all k . We apply the Basic Strategy to solve this system.

Theorem 6. *Let a QCG be defined by $F(x) = ax^2 + b$ over $\mathbb{Z}_p$ with known a and b . Given $n+1$ consecutive truncated outputs for fixed $n \geq 1$, the entire sequence can be predicted, provided that Assumption 1 holds and that the single continuous unknown part of size p^δ for each iteration satisfies*

$$\delta < \frac{n}{2n+1} - \varepsilon,$$

for some negligible ε . The time complexity is polynomial in $\log p$. As $n \rightarrow \infty$, the asymptotic bound is $\delta < 1/2$.

Proof. The proof is analogous to that of Theorem 3. The leading monomial of each derived polynomial f_j is x_j^2 . Let m be a large positive integer, and define the same elementary vectors $\alpha_j^{(n+1)} \in \mathbb{Z}^{n+1}$ as (7). The sets A and B are defined as

$$\begin{aligned} A &= \{\mathbf{0}, \alpha_1^{(n+1)}, \dots, \alpha_{n+1}^{(n+1)}, 2\alpha_1^{(n+1)}, \dots, 2\alpha_n^{(n+1)}\} \subseteq \mathbb{Z}^{n+1}, \\ B &= \{2\alpha_1^{(n+1)}, \dots, 2\alpha_n^{(n+1)}\} \subseteq \mathbb{Z}^{n+1}. \end{aligned}$$

Similar to the proof of Theorem 3, we use A and B to construct the shift polynomials $g_\lambda(x_1, \dots, x_{n+1})$ for each monomial λ , and form the lattice Λ using the coefficient vectors of $g_\lambda(x_1 X_1, \dots, x_{n+1} X_{n+1})$ as its basis. Observe that the basis matrix of Λ is diagonal. Hence, the solvable condition from (6) becomes

$$\prod_{j=1}^{n+1} X_j^{\sigma_j} p^{\sigma_p} < p^{\sigma_m - \varepsilon_0}.$$

It remains to compute the values of σ_m , $\sigma_1, \dots, \sigma_{n+1}$, and σ_p .

Computing σ_m . To evaluate σ_m , we denote by N the convex hull of A and apply the asymptotic estimate from Lemma 3:

$$\lim_{m \rightarrow \infty} \frac{\sigma_m}{m^{n+2}} = \lim_{m \rightarrow \infty} \frac{|mA|}{m^{n+1}} = \int_N 1 \, dV.$$

Let N_0 denote the convex hull of the points $\{\mathbf{0}, \alpha_1^{(n+1)}, \dots, \alpha_{n+1}^{(n+1)}\}$. It follows that

$$\int_N 1 \, dV = 2^n \cdot \int_{N_0} 1 \, dV = \frac{2^n}{(n+1)!}.$$

Therefore, we obtain

$$\sigma_m = \frac{2^n}{(n+1)!} m^{n+2} + o(m^{n+2}).$$

Computing σ_j . The asymptotic relation given by Lemma 4 can be expressed as

$$\lim_{m \rightarrow \infty} \frac{\sigma_j}{m^{n+2}} = \int_N x_j dV = \begin{cases} 2^{n+1} \int_{N_0} x_j dV & \text{for } j = 1, \dots, n, \\ 2^n \int_{N_0} x_j dV & \text{for } j = n+1. \end{cases}$$

Similarly, we conclude that

$$\sigma_1 = \sigma_2 = \dots = \sigma_n = 2\sigma_{n+1} = \frac{2^{n+1}}{(n+2)!} m^{n+2} + o(m^{n+2}).$$

Computing σ_p . We first define $\tilde{A}$ as

$$\begin{aligned} \tilde{A} = \{ & (i_1, \dots, i_{n+1}, 1) \mid (i_1, \dots, i_{n+1}) \in A \} \\ & \cup \{ (i_1, \dots, i_{n+1}, 0) \mid (i_1, \dots, i_{n+1}) \in B \} \subseteq \mathbb{Z}^{n+2}, \end{aligned}$$

and let $\tilde{N}$ be the convex hull of $\tilde{A}$. Hence, we have

$$\lim_{m \rightarrow \infty} \frac{\sigma_p}{m^{n+2}} = \int_N 1 dV - \int_{\tilde{N}} 1 dV.$$

Then we define the same $\tilde{N}_0$ as in Theorem 3. Accordingly, we have

$$\int_{\tilde{N}} 1 dV = 2^n \cdot \int_{\tilde{N}_0} 1 dV = \frac{2^{n+1}}{(n+2)!}$$

Therefore, we conclude that

$$\sigma_p = \frac{2^{n+1}}{(n+2)!} m^{n+2} + o(m^{n+2}).$$

Plugging these into the solvable condition with $X_j = p^\delta$ gives

$$\delta \left(n \cdot \frac{2^{n+1}}{(n+2)!} + \frac{2^n}{(n+2)!} \right) + \frac{2^{n+1}}{(n+2)!} < \frac{2^n}{(n+1)!} - \varepsilon_1,$$

which simplifies to

$$\delta(2n+1) + 2 < (n+2) - \varepsilon_2,$$

leading to the bound

$$\delta < \frac{n}{2n+1} - \varepsilon.$$

The time complexity is polynomial in $\log p$. □

A.2 QCG with Unknown a, b

A QCG is defined by the recurrence $v_{i+1} = av_i^2 + b \bmod p$. We consider the case where the parameters a, b are unknown. Given $n + 1$ consecutive outputs, we have n relations $v_j = av_{j-1}^2 + b \bmod p$. We shall eliminate b first by subtracting consecutive relations:

$$v_j - v_{j+1} = av_{j-1}^2 - av_j^2 \bmod p,$$

which can be rearranged to

$$a \equiv \frac{v_j - v_{j+1}}{v_{j-1}^2 - v_j^2} \bmod p.$$

Then we further eliminate a to obtain a modular equation $f_j(x_j, x_{j+1}, x_{j+2}, x_{j+3})$ in four variables, whose leading monomial is x_{j+1}^3 in the graded reverse lexicographic order. We substitute $v_k = w_k + L_k u_k$ and let $x_k = u_{k-1}$ to obtain a system of $n - 2$ modular equations of the form

$$\begin{cases} f_1(x_1, x_2, x_3, x_4) \equiv 0 \bmod p, \\ f_2(x_2, x_3, x_4, x_5) \equiv 0 \bmod p, \\ \vdots \\ f_{n-2}(x_{n-2}, x_{n-1}, x_n, x_{n+1}) \equiv 0 \bmod p. \end{cases}$$

The root of interest is $(x_1^*, \dots, x_{n+1}^*) = (u_0, \dots, u_n)$, with bounds $X_k = p^\delta$ for all k . We apply the Basic Strategy. The analysis is analogous to that in Section 3.2, but with different Newton polytopes reflecting the cubic leading terms.

Theorem 7. *Let a QCG be defined by $F(x) = ax^2 + b$ over $\mathbb{Z}_p$ with unknown a and b . Given $n + 1$ consecutive truncated outputs for fixed $n \geq 3$, the entire sequence can be predicted, provided that Assumption 1 holds and that the single continuous unknown part of size p^δ for each iteration satisfies*

$$\delta < \frac{n-2}{3n+3} - \varepsilon,$$

for some negligible ε . The time complexity is polynomial in $\log p$. As $n \rightarrow \infty$, the asymptotic bound is $\delta < 1/3$.

Proof. As the leading monomial of each f_j in the graded reverse lexicographic order is x_{j+1}^3 . We define the sets A and B are defined as

$$\begin{aligned} A &= \{\mathbf{0}, \alpha_1^{(n+1)}, \dots, \alpha_{n+1}^{(n+1)}, 3\alpha_1^{(n+1)}, \dots, 3\alpha_{n+1}^{(n+1)}\} \subseteq \mathbb{Z}^{n+1}, \\ B &= \{3\alpha_2^{(n+1)}, \dots, 3\alpha_{n-1}^{(n+1)}\} \subseteq \mathbb{Z}^{n+1}. \end{aligned}$$

Similar to the proof of Theorem 2, we include $3\alpha_1^{(n+1)}$ and $3\alpha_{n+1}^{(n+1)}$ in A to simplify the volume calculations, as the polynomials f_j have degree 3 in all

variables. We use A and B to construct the shift polynomials $g_\lambda(x_1, \dots, x_{n+1})$ for each monomial λ , and form the lattice Λ using the coefficient vectors of $g_\lambda(x_1 X_1, \dots, x_{n+1} X_{n+1})$ as its basis. Observe that the basis matrix of Λ is diagonal. Hence, the solvable condition from (6) becomes

$$\prod_{j=1}^{n+1} X_j^{\sigma_j} p^{\sigma_p} < p^{\sigma_m - \varepsilon_0}.$$

It remains to compute the values of $\sigma_m, \sigma_1, \dots, \sigma_{n+1}$, and σ_p .

Computing σ_m . To evaluate σ_m , we denote by N the convex hull of A and apply the asymptotic estimate from Lemma 3:

$$\lim_{m \rightarrow \infty} \frac{\sigma_m}{m^{n+2}} = \lim_{m \rightarrow \infty} \frac{|mA|}{m^{n+1}} = \int_N 1 \, dV = \frac{3^{n+1}}{(n+1)!}.$$

Therefore, we obtain

$$\sigma_m = \frac{3^{n+1}}{(n+1)!} m^{n+2} + o(m^{n+2}).$$

Computing σ_j . For each σ_j , we apply Lemma 4, which gives

$$\lim_{m \rightarrow \infty} \frac{\sigma_j}{m^{n+2}} = \int_N x_j \, dV = \frac{3^{n+2}}{(n+2)!}.$$

Hence,

$$\sigma_1 = \dots = \sigma_{n+1} = \frac{3^{n+2}}{(n+2)!} m^{n+2} + o(m^{n+2}).$$

Computing σ_p . To compute σ_p , we define $\tilde{A}$ as follows:

$$\begin{aligned} \tilde{A} = & \{(i_1, \dots, i_{n+1}, 1) \mid (i_1, \dots, i_{n+1}) \in A\} \\ & \cup \{(i_1, \dots, i_{n+1}, 0) \mid (i_1, \dots, i_{n+1}) \in B\} \subseteq \mathbb{Z}^{n+2}. \end{aligned}$$

Let $\tilde{N}$ be the convex hull of $\tilde{A}$. Then, we have

$$\lim_{m \rightarrow \infty} \frac{\sigma_p}{m^{n+2}} = \int_N 1 \, dV - \int_{\tilde{N}} 1 \, dV.$$

Our task reduces to evaluating $\int_{\tilde{N}} 1 \, dV$. Similarly, we introduce a simplified polytope $\tilde{N}_0$, defined as the convex hull of the following point set

$$\begin{aligned} \tilde{A}_0 = & \{(i_1, \dots, i_{n+1}, 1) \mid (i_1, \dots, i_{n+1}) \in \{\mathbf{0}, \alpha_1^{(n+1)}, \dots, \alpha_{n+1}^{(n+1)}\}\} \\ & \cup \{(i_1, \dots, i_{n+1}, 0) \mid (i_1, \dots, i_{n+1}) \in \{\alpha_2^{(n+1)}, \dots, \alpha_{n-1}^{(n+1)}\}\} \subseteq \mathbb{Z}^{n+2}. \end{aligned}$$

It implies that

$$\int_{\tilde{N}} 1 \, dV = 3^{n+1} \cdot \int_{\tilde{N}_0} 1 \, dV.$$

Following the same idea used in Theorem 1 and Theorem 2, let S denote the convex hull of $\tilde{A}_0 \cup \{\mathbf{0}\} \subseteq \mathbb{Z}^{n+2}$, and define T as the simplex generated by the points

$$\{\mathbf{0}, \alpha_2^{(n+2)}, \dots, \alpha_{n-1}^{(n+2)}, \alpha_{n+2}^{(n+2)}, (1, 0, \dots, 0, 1), (0, \dots, 0, 1, 0, 1), (0, \dots, 0, 1, 1)\} \subseteq \mathbb{Z}^{n+2},$$

which has volume $1/(n+2)!$. Then, we have

$$\int_{\tilde{N}_0} 1 \, dV = \int_S 1 \, dV - \int_T 1 \, dV.$$

For the computation of $\int_S 1 \, dV$, We define the projection map ψ by taking the components from the second to the $(n-1)$ -st together with the $(n+2)$ -nd component, that is,

$$\begin{aligned} \psi : \mathbb{Z}^{n+2} &\longrightarrow \mathbb{Z}^{n-1}, \\ (i_1, \dots, i_{n+2}) &\longmapsto (i_2, \dots, i_{n-1}, i_{n+2}), \end{aligned}$$

$\psi(S)$ forms an $n-2$ -dimensional prism of height 1 with base that is the polytope generated by

$$\{\mathbf{0}, \alpha_1^{(n-2)}, \dots, \alpha_{n-2}^{(n-2)}\} \subseteq \mathbb{Z}^{n-2}.$$

Thus, the volume of $\psi(S)$ remains $1/(n-2)!$ and we have

$$\begin{aligned} \int_S 1 \, dV &= \int_{x_{n+1}=0}^1 \int_{x_n=0}^{1-x_{n+1}} \int_{x_1=0}^{1-x_n-x_{n+1}} \frac{(1-x_1-x_n-x_{n+1})^{n-1}}{(n-2)!} \, dx_1 \, dx_n \, dx_{n+1} \\ &= \frac{n-1}{(n+2)!}. \end{aligned}$$

Combining the above results, we conclude that

$$\sigma_p = \frac{4 \cdot 3^{n+1}}{(n+2)!} m^{n+2} + o(m^{n+2}).$$

Plugging these into the solvable condition gives

$$\delta \cdot (n+1) \cdot \frac{3^{n+2}}{(n+2)!} + \frac{4 \cdot 3^{n+1}}{(n+2)!} < \frac{3^{n+1}}{(n+1)!} - \varepsilon_1.$$

Multiplying by $(n+2)!$ and dividing by 3^{n+1} simplifies this to

$$\delta \cdot 3(n+1) + 4 < (n+2) - \varepsilon_2,$$

which yields the bound

$$\delta < \frac{n-2}{3n+3} - \varepsilon.$$

The complexity analysis is similar to that of Theorem 1. Therefore, the time complexity is polynomial in $\log p$. $\square$

A.3 Pollard Generator with Known c

A Pollard Generator is defined by the recurrence $v_{i+1} = v_i^2 + c \bmod p$. We consider the case where the parameter c is known. Given $n + 1$ consecutive outputs, we have n relations $v_j = v_{j-1}^2 + c \bmod p$. There is no need to eliminate c , and we directly get the consecutive relations:

$$v_{j-1}^2 - v_j + c \equiv 0 \bmod p.$$

This is the same relation as that derived for QCGs with known $a = 1$ and known b . We substitute $v_k = w_k + L_k u_k$ and let $x_k = u_{k-1}$ to obtain a monic modular equations of the form

$$f_j(x_j, x_{j+1}) = x_j^2 + c_{j1}x_j + c_{j2}x_{j+1} + c_{j0} \equiv 0 \bmod p.$$

Hence, the system of n modular equations to be solved is

$$\begin{cases} f_1(x_1, x_2) \equiv 0 \bmod p, \\ f_2(x_2, x_3) \equiv 0 \bmod p, \\ \vdots \\ f_n(x_n, x_{n+1}) \equiv 0 \bmod p. \end{cases}$$

The root of interest is $(x_1^*, \dots, x_{n+1}^*) = (u_0, \dots, u_n)$, with bounds $X_k = p^\delta$ for all k . We apply the Basic Strategy to solve this system.

Theorem 8. *Let a Pollard generator be defined by $F(x) = x^2 + c$ over $\mathbb{Z}_p$ with known c . Given $n + 1$ consecutive truncated outputs for fixed $n \geq 1$, the entire sequence can be predicted, provided that Assumption 1 holds and that the single continuous unknown part of size p^δ for each iteration satisfies*

$$\delta < \frac{n}{2n+1} - \varepsilon,$$

for some negligible $\varepsilon > 0$. The time complexity is polynomial in $\log p$. As $n \rightarrow \infty$, the asymptotic bound is $\delta < 1/2$.

Proof. This can be viewed as a special case of $a = 1$ for QCGs with known a and b . See the proof of Theorem 6. $\square$